THE MASTERMIND COACHING CLASSES

Revision Notes

Class 6 · Science

NCERT Curiosity — Chapters 7 to 12 · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
7	Temperature and its Measurement	Hotness, thermometers, Celsius/Fahrenheit/Kelvin, correct reading
8	A Journey through States of Water	States of water, evaporation, condensation, water cycle
9	Methods of Separation in Everyday Life	Handpicking, threshing, winnowing, sieving, evaporation, filtration, magnetic separation
10	Living Creatures: Exploring their Characteristics	Living vs non-living, seed germination, life cycles
11	Nature's Treasures	Air, water, Sun, forests, soil, rocks, fossil fuels; renewable and non-renewable resources
12	Beyond Earth	Stars, constellations, night sky, Solar System, Moon, galaxy

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CHAPTER 07

Temperature and its Measurement

*Lambok in Shillong feels feverish; his sister Phiban checks with a thermometer instead of just touching his forehead. The chapter explains why our sense of touch is unreliable, defines **temperature**, introduces clinical and laboratory thermometers, temperature scales and the correct way to use and read a thermometer. The opening quote is by Anna Mani: wrong measurements are worse than no measurements at all.*

Big Questions

- Can we judge hot and cold by touch?
- What is temperature and how is it measured?
- How do we use a thermometer correctly?
- What are the temperature scales?

7.1 Hot or Cold?

Activity 7.1: Keep three containers: A (warm water), B (tap water), C (ice-cold water). Put the right hand in A and the left hand in C for 1–2 minutes, then both hands together in B. The right hand feels B **cool** while the left hand feels the same water **warm**.

IMPORTANT

Our sense of touch is not reliable for judging how hot or cold a body is; we need a measuring device.

7.2 Temperature

DEFINITION

Temperature is a reliable measure of the **hotness or coldness** of a body. A hotter body has a higher temperature. A device that measures temperature is a **thermometer**.

Two common kinds: **clinical thermometers** (human body temperature) and **laboratory thermometers** (other purposes).

7.3 Measuring Temperature

▸ 7.3.1 Clinical thermometer

- Modern ones are **digital**, battery-run and use **heat sensors**; they measure when placed in contact with the body (under the tongue).
- Older **mercury thermometers** are being replaced because mercury is **toxic** and hard to dispose of if the thermometer breaks; digital ones are safer and easier to read.
- During COVID-19, **non-contact (infrared) thermometers** measured temperature without touching, reducing the spread of disease.

▸ Precautions while using a digital clinical thermometer

Read the manual
Use it only after reading the instruction manual.

Clean
Wash hands and wash the thermometer tip with soap and water before and after use; keep the display out of water; do not hold it by the tip.

Reset
Press the reset button.

Place
Put under the tongue and close the mouth.

Read
Wait for the beep or light, take it out and read the display; record and clean it again.

- **Normal human body temperature = 37.0 °C (98.6 °F)**, the **average** of many healthy people; a healthy person may differ slightly. It varies with **age, time of day and activity level**.
- Human body temperature normally stays between **35 °C and 42 °C**.
- For small children or old people, the thermometer may be placed in the **armpit**; the reading is **0.5 °C to 1 °C lower** than the actual body temperature.
- Pulse rate was used in olden India to detect fever but is **not a reliable indicator** as other situations also affect it.
- A clinical thermometer **cannot** measure boiling water or ice, which lie outside its range.

▸ Temperature scales

Scale	Unit and symbol	Notes
Celsius	degree Celsius, °C	Used in clinical thermometers; commonly used
Fahrenheit	degree Fahrenheit, °F	37.0 °C = 98.6 °F; no longer used in most scientific work
Kelvin	kelvin, K (no degree sign)	**SI unit of temperature**

FORMULA
Temperature in kelvin = Temperature in °C + 273.15

NOTE
Scale names (Celsius, Fahrenheit, Kelvin) start with capital letters; 'degree' and 'kelvin' are lowercase; symbols (°C, °F, K) are capitals with **no full stop** and a **space** between number and unit; use 'degrees' for more than one degree.

▸ 7.3.2 Laboratory thermometer

A long, narrow, uniform, sealed **glass tube** with a **bulb** at one end containing a liquid (**alcohol coloured red, or mercury**) and a Celsius scale along the tube. The liquid column **rises or falls** with temperature; the mark level with the top of the column is the reading. Typical **range: –10 °C to 110 °C**.

Term	Meaning	Example
Range	Lowest to highest temperature it can measure	–10 °C to 110 °C
Least count (smallest value)	Temperature difference between two big marks ÷ number of small divisions between them	10 °C ÷ 10 = **1 °C**

Always check the range and smallest division of the thermometer before use.

▸ Correct way of using a laboratory thermometer

- The **bulb must not touch** the bottom or sides of the beaker.
- Hold the thermometer **vertically**, not tilted.
- Read the temperature **while the bulb is still in the water** (the column falls as soon as it is taken out).
- Keep the **eye in line** with the top of the liquid column.
- Wait until the column **stops rising**, but not so long that water cools.
- Handle with care (glass breaks) and do not hold by the bulb.

Differences in readings among students (e.g. boiling water at Shillong: 97.8 to 98.1 °C) show that **correct reading procedure was not followed by all**. Note: **temperature stays constant while ice melts and while water boils.** Experiments with boiling water are done only under the teacher's supervision.

▸ 7.3.3 Air temperature

- Wall (room) thermometers show approximate room temperature; weather reports give **maximum and minimum air temperatures** of the day.
- They vary from day to day; rise towards summer and fall towards winter.
- Air temperature is monitored at weather stations all over the world and used in weather forecasts.

▸ Know a scientist — Anna Mani (1918–2001)

The **'Weather Woman of India'**; invented and built many weather-measurement instruments, reducing India's dependence on other countries, and explored wind and solar energy possibilities, helping India become a leader in renewable energy.

> **REMEMBER**
> The Sun's core reaches about 15 million °C. There is no upper limit on temperature, but the lowest possible temperature is close to **–273.15 °C (0 K), called absolute zero**.

7.4 Likely Exam Questions

Likely question	Model answer
Why can't we rely on touch to judge hotness?	The hands adjust to earlier temperature, so the same water feels cool to one hand and warm to the other.
Define temperature.	A reliable measure of how hot or cold a body is.
Name two types of thermometers and their uses.	Clinical – body temperature; laboratory – other measurements.
Why are digital thermometers replacing mercury ones?	Mercury is toxic and hard to dispose of; digital ones are safe and easy to read.
What is the normal human body temperature?	37.0 °C or 98.6 °F.
Why can't a clinical thermometer measure boiling water?	Its range (35–42 °C) is far below 100 °C.
What is the SI unit of temperature? Convert 27 °C to it.	Kelvin; 27 + 273.15 = 300.15 K.
State precautions for using a laboratory thermometer.	Bulb not touching sides/bottom, vertical, read in liquid, eye in line, careful handling.

Likely question	Model answer
How do you find the smallest value of a thermometer?	Divide difference between two big marks by the number of small divisions.
A thermometer has 50 divisions between 0 °C and 100 °C. What does one division read?	100 ÷ 50 = 2 °C.
Why should the thermometer be read while in water?	The liquid column falls as soon as it is taken out.
What is absolute zero?	The lowest possible temperature, about –273.15 °C (0 K).

★ KEY TAKEAWAYS

- Touch is unreliable; temperature measures hotness objectively.
- Clinical thermometers read body temperature (35–42 °C); normal is 37.0 °C = 98.6 °F.
- Laboratory thermometer range is usually –10 °C to 110 °C.
- SI unit is kelvin; K = °C + 273.15.
- Use a thermometer vertically, bulb not touching the container, eye level with liquid, read while immersed.
- Temperature of melting ice and boiling water stays constant during the change.

CHAPTER 08

A Journey through States of Water

*Aavi and Thirav, looking at ice in their lemonade, ask whether ice and water are the same thing. Through puddles, steel plates, tumblers of ice water and clay pots they discover that water exists as **solid, liquid and gas**, changes state on heating or cooling, **evaporates** and **condenses**, and travels through the **water cycle**. The chapter also trains the habit of observing, questioning, reasoning and testing.*

Big Questions

- Are ice, water and vapour the same substance?
- Where does water go when it disappears, and where do droplets on a cold glass come from?
- What affects the speed of evaporation?
- How do clouds and rain form?

8.0 Ice and Water Are One Substance

Activity 8.1: an ice cube left in a cup turns into water, so **ice and water are two forms (states) of the same substance**. Their behaviour differs — water flows and splashes, ice does not.

8.1 Investigating Water's Disappearing Act

- Puddles on the playground and water on washed utensils disappear. Aavi suggests water **seeps** into the ground/surface; Thirav disagrees.
- Activity 8.2: a tablespoon of water on a **steel plate** does not seep through, yet vanishes — it has turned into **water vapour**, the gaseous state of water.
- Steam from water sprinkled on a hot dosa pan is actually water vapour (invisible); visible 'steam' is tiny water droplets formed from some vapour.

DEFINITION

Evaporation is the conversion of water (liquid) into water vapour. It takes place **continuously, even at room temperature** (wet clothes drying, mopped floors, sweat, hand sanitiser disappearing).

Puddles disappear by evaporation — and partly by seeping into soil (both).

8.2 Another Mystery: Droplets on the Glass

Activity 8.3: cold water with ice in a glass tumbler forms **water droplets on the outer surface** after a few minutes; small drops merge into bigger ones. Possible reasons: (a) ice came out, (b) water seeped out through glass, (c) water vapour from air condensed. Reasoning: water level did not fall, so seeping is unlikely.

DEFINITION

Condensation is the conversion of water vapour into liquid water when vapour in air touches a **cold surface** (dew on plants in the morning; drops on the inside of a lid covering boiling water).

Activity 8.4: a tumbler of ice water covered with a steel plate is weighed every 5 minutes for 30 minutes — the **mass increases**, because water vapour from the air condenses on the tumbler. Marking the water level shows

it **does not go down**, proving no seepage; extra water is from condensation. The amount of water vapour in air is called **humidity**.

8.3 States of Water

Property	Ice (solid)	Water (liquid)	Water vapour (gas)
Shape	Fixed; keeps its shape in any container	No fixed shape; takes shape of container	No fixed shape
Flow	Does not flow	Flows	Spreads
Spreading	Does not spread	Spreads on a surface; volume stays constant	Spreads throughout available space
Visibility	Visible	Visible	Invisible, present in air even at room temperature

- Other substances also show three states: wax, oil, ghee.
- Solids: stones, wood, glass. Liquids: milk, oil. Gases: oxygen, carbon dioxide.
- The smell of cooking food reaches us because **gases spread** through air.

8.4 Changing the States of Water

Process	Change	Condition
Melting	Solid → liquid (ice → water)	Supply heat
Freezing	Liquid → solid (water → ice)	Cool, e.g. in a freezer
Evaporation	Liquid → gas (vapour)	Heating or at room temperature
Condensation	Gas → liquid	Cooling / contact with a cold surface

- Candle wax melts on heating and solidifies on cooling; coconut oil solidifies in winter.
- Atmospheric Water Generators collect drinking water by cooling humid air so vapour condenses.

8.5 Faster or Slower Evaporation

In investigations change **one condition** and keep others the same.

Factor	Effect on evaporation	Everyday example
Exposed area (plate vs bottle cap)	More area → **faster**	Spread clothes out to dry
Temperature / sunlight	Hotter / sunny → **faster**	Clothes dry quickly on hot days
Air movement (wind)	More wind → **faster**	Clothes dry faster on a windy day or under a fan
Humidity	High humidity → **slower**	Clothes dry slowly on rainy days

8.6 Cooling Effect

- Water in a **matka** (earthen pot) is cold because water seeps through tiny pores and **evaporates from the surface, cooling the water** inside. A stainless steel pot has no pores. The **surahi** works the same way.

- Other examples: sprinkling water on a floor or roof in summer, sweat evaporating, sanitiser feeling cool, sitting under a fan (moving air helps sweat evaporate).
- **Pot-in-pot cooler** (Activity 8.9): small earthen pot inside a larger one, gap filled with sand, sand kept moist, covered with a lid or wet jute sack; evaporation from wet sand cools the inner pot in 4–5 hours and keeps vegetables and fruits fresh for days.

> REMEMBER
> **Evaporation causes cooling.**

8.7 How Do Clouds Give Us Rain?

Evaporation
Water from oceans and the Earth's surface becomes water vapour, which is lighter than air and rises.

Cooling
Higher up the air gets cooler and cooler.

Condensation
Vapour condenses into tiny droplets around **dust particles**; these floating droplets form **clouds**.

Growth
Many droplets join to form bigger drops.

Precipitation
Heavy drops fall as **rain**, or as **hail or snow** under special conditions.

Activity 8.10: in a bottle with a little water, squeeze and release repeatedly. Haze (cloud) appears much more clearly when a **burnt piece of newspaper** is added — it provides dust particles around which vapour condenses. (Handle burning paper carefully.)

▸ The water cycle

> DEFINITION
> The continuous circulation of water between Earth's surface and atmosphere — evaporation from ocean/land, condensation to clouds, rain/snow/hail, flow through rivers, lakes and groundwater back to oceans — is the **water cycle**.

Only a small portion of Earth's water is fit for use by plants, animals and humans; most is in oceans. With rising population, demand grows and shortages occur, so **use water wisely and keep water bodies unpolluted**.

8.8 Likely Exam Questions

Likely question	Model answer
Are ice, water and water vapour different substances?	No, they are three states of the same substance, water.
Define evaporation and give two examples.	Conversion of liquid water into vapour; drying clothes, disappearing puddles.
Define condensation. Give an example.	Water vapour turning to liquid on cooling; drops on a cold glass, dew.
Why do droplets form on the outside of a glass of cold water?	Water vapour in air condenses on its cold surface; water does not seep through glass.

Likely question	Model answer
What is humidity?	The amount of water vapour present in air.
Compare solid, liquid and gas states of water.	Solid has fixed shape; liquid takes container's shape and flows; gas spreads in all available space.
What factors affect the rate of evaporation?	Exposed area, temperature, wind and humidity.
Why do clothes dry slowly on rainy days?	High humidity makes evaporation slow.
Why is water in a matka cool?	Water seeping through pores evaporates and cools the water.
How are clouds formed?	Vapour rises, cools and condenses around dust particles to form tiny droplets.
What is the water cycle?	Continuous circulation of water between Earth and atmosphere through evaporation, condensation and precipitation.
How can a hot two-wheeler seat be cooled?	Sprinkle water on it so evaporation cools it, or keep it in shade.

★ KEY TAKEAWAYS

- Water exists as ice (solid), water (liquid) and vapour (gas).
- Evaporation: liquid → vapour; condensation: vapour → liquid; melting and freezing link solid and liquid.
- Evaporation is faster with larger area, higher temperature and wind, and slower with high humidity.
- Evaporation causes cooling (matka, pot-in-pot cooler, sweat).
- Clouds form by condensation on dust particles; rain, hail or snow follow.
- The water cycle circulates water; conserve it and avoid pollution.

CHAPTER 09

Methods of Separation in Everyday Life

*Malli and Valli's 'Bharat ki Yatra' across India shows how everyday life depends on **separating mixtures**: grains from stones, husk from grain, bran from flour, salt from seawater, tea leaves from tea, butter from curd and iron nails from sawdust. Kabir's couplet compares winnowing — husk blown away, grain retained — to keeping virtues and discarding vices.*

Big Questions

- Why do we separate substances from a mixture?
- Which method suits which kind of mixture?
- How can a solid dissolved in a liquid be recovered?
- How are mixtures with many components separated?

9.1 Why Do We Separate Substances?

We separate to **remove unwanted/harmful components** (stones from grain, tea leaves from tea) and to **obtain a useful component** (butter from curd, salt from seawater). The right method depends on the **properties of the components** — size, weight, solubility, magnetism, etc.

9.2 Methods of Separation

▸ (a) Handpicking

Picking particles by hand from a mixture on the basis of differences in **size, colour and shape** (stones and husk from wheat or rice; black pepper from pulao). Convenient when the quantity to be removed is **small** and easily picked.

▸ (b) Threshing

Separating **grains from harvested stalks** by beating them (on a wooden log); mechanical **threshers** do threshing and winnowing together.

▸ (c) Winnowing

Separating **heavier and lighter components** using wind or blown air. Farmers pour threshed grain from a raised platform in the direction of the wind, using a **soop (bamboo tray)**: heavy grains fall near, lighter husk blows away. Similarly, blowing on rubbed roasted peanuts removes the light skins.

▸ (d) Sieving

Used when components of a **solid–solid mixture differ in size**. Fine flour passes through holes; bran and small stones stay on the sieve. The holes must be smaller than the particles to be retained and larger than those to pass. Also used at construction sites to separate pebbles/stones from sand.

▸ (e) Evaporation

- Used to recover a **solid dissolved in a liquid**. Water evaporates, leaving solid behind.
- **Salt from seawater:** seawater is kept in shallow pits in sunlight and air; water evaporates over days leaving a solid mixture, from which common salt is purified. Sources: seawater, **Sambhar Lake (Rajasthan)**. Namak Satyagraha (**Dandi March**) is associated with Sabarmati Ashram, Ahmedabad.

- Activity 9.2: drops of salt solution on dark paper dry, leaving white salt patches (like white patches on clothes after sweating). Activity 9.3 (teacher demo): heating salt solution in a **china dish** leaves salt behind.
- Ayurvedic herbs are shade-dried so excess water evaporates, leaving the medicinal parts.
- Evaporation alone does not give back the water; to recover both salt and water, evaporated water vapour must be **condensed**.

▸ (f) Sedimentation and decantation

Term	Meaning	Example
Sedimentation	Settling of heavier insoluble component at the bottom of a liquid	Tea leaves settle in a pan; mud in water; rice and pulses before washing
Decantation	Removing the liquid by gently tilting the vessel, leaving the sediment	Pouring tea off leaves; draining water from washed rice; oil from water

Decantation does **not completely** separate (a few tea leaves may pour out), so it is a rough method. Oil and water separate into layers on standing, then can be decanted.

▸ (g) Filtration

Separating **insoluble solid particles from a liquid** by passing the mixture through a **filter** that has pores. Tea strainer for tea leaves; cloth pores (used since ancient times) for muddy water; **filter paper** (finest pores) folded into a cone in a funnel over a conical flask (Activity 9.4).

Term	Meaning
Residue	Solid left on the filter paper (mud)
Filtrate	Clear liquid collected in the flask

- Cotton, charcoal and sand can also act as filters; the choice depends on the **size of particles** to be removed.
- Tea bags were first of silk, then gauze/muslin, now filter paper.
- A fisherman's **net** acts like a filter — water drains through the mesh while fish are retained (but also plastic waste, showing river pollution: stop plastic at source).
- Nasal hair and mask material also filter dust.

▸ (h) Churning

Curd or milk is churned with a **mathni (churner)** or machine; lighter **butter floats on top** and buttermilk (chhach) remains. Used to extract butter.

▸ (i) Magnetic separation

Separating **magnetic from non-magnetic substances with a magnet** — iron nails from sawdust; recyclers separate scrap iron from waste using magnets fitted on cranes.

9.3 Summary Table

Mixture	Method	Property used
Stones in pulses; black pepper in pulao	Handpicking	Size, colour, shape
Grains and stalks	Threshing	Beating stalks

Mixture	Method	Property used
Husk and grains	Winnowing	Weight (wind)
Flour and bran	Sieving	Particle size
Salt from salt solution/seawater	Evaporation	Water evaporates, salt does not
Heavy insoluble solid and liquid	Sedimentation + decantation	Settles down
Muddy water; tea leaves	Filtration	Insoluble particles larger than pores
Butter from curd	Churning	Lighter butter floats
Iron nails and sawdust	Magnetic separation	Magnetic property

IMPORTANT

A mixture with **more than two components needs a combination of methods**. Example: iron nails, sand, pepper, stones, salt and water — first magnet (nails), handpick pepper/stones, then add water to dissolve salt, filter off sand, and evaporate to get salt. Plan the sequence from the property that is easiest to use.

9.4 Likely Exam Questions

Likely question	Model answer
Why do we separate components of a mixture?	To remove unwanted components or to obtain useful ones.
Define handpicking. When is it suitable?	Picking by hand on basis of size, colour, shape; when impurity quantity is small.
Differentiate threshing and winnowing.	Threshing separates grain from stalks by beating; winnowing separates husk from grains using wind.
What is sieving?	Separating solid components by particle size using a sieve.
How is salt obtained from seawater?	Seawater is evaporated in shallow pits by sunlight, then the salt is purified.
What are sedimentation and decantation?	Settling of heavier insoluble solid; pouring off the liquid by tilting.
Why is decantation not a proper method of separation?	It cannot remove all the particles.
What is filtration? Define residue and filtrate.	Separating insoluble solid from liquid with a filter; solid left is residue and the liquid collected is filtrate.
Which method separates butter from curd?	Churning.
How can iron nails be separated from sawdust?	By magnetic separation with a magnet.
Which method is used to separate oil from water?	Decantation (after layers form).
Outline separating potato, salt and sawdust.	Handpick potatoes; add water, sawdust floats/skim or filter; filter and evaporate the filtrate to get salt.

★ KEY TAKEAWAYS

- Separation removes unwanted or obtains useful components; method depends on property differences.
- Handpicking (size/colour), threshing, winnowing (weight), sieving (size).
- Evaporation recovers dissolved solids; sedimentation + decantation and filtration separate insoluble solids from liquids.
- Churning separates butter; magnets separate iron.
- Combine several methods for multi-component mixtures.

CHAPTER 10

Living Creatures: Exploring their Characteristics

*Avadhi and Aayush find a snail shell and learn it is part of a living animal. This leads to the question: **what sets the living apart from the non-living?** The chapter lists the characteristics of life, investigates conditions needed for seed germination and direction of root/shoot growth, and follows the life cycles of a plant, mosquito and frog.*

Big Questions

- What are the characteristics of living beings?
- What do seeds need to germinate?
- How do roots and shoots grow?
- What are the life cycles of a mosquito and a frog?

10.1 What Sets the Living Apart from the Non-living?

Compare things with yourself. A car moves but is non-living, so movement alone is not enough; a car does not grow, eat, breathe, reproduce — we use **a set of characteristics** and not just one.

Characteristic	Meaning / explanation	Examples
Movement	Animals move from place to place; plants show movement without changing place	Opening of flowers; **Drosera** (insectivorous plant: saucer-shaped leaves with sticky-tipped hairs that fold inward to trap insects); climbers winding around supports
Growth	Increase in size, needs food (nutrition)	A child outgrows old clothes; plants and all living beings grow
Nutrition	Living beings need food for growth and energy	All living beings
Respiration	Breathing (inhale/exhale) is part of respiration; breathing rate rises after exercise	Animals breathe (abdomen movement in dogs, cows); plants take air in and out through tiny pores called **stomata** on leaves
Excretion	Removal of waste products from the body	Sweat (water and salts), urine; plants give out excess water and minerals as droplets on leaves of grass and roses
Response to stimuli	A **stimulus** is anything or any event that prompts a living being to respond	Moving away from a thorn or hot cup; **touch-me-not (mimosa / chhui-mui / lajjalu)** folds leaves on touch; amla leaves fold after sunset
Reproduction	Producing new ones of one's own kind; ensures **continuity of life**	Young ones of cats, dogs; plants produce seeds. Non-living things (pencil, chair) have no young ones
Death	A living being that cannot show these characteristics even when food, air and water are available is dead	—

IMPORTANT

All living beings **move, need food, grow, respire, excrete, respond to stimuli, reproduce and eventually die**. Absence of these features indicates a non-living thing. A seed is considered living because it can germinate and grow.

On the snail: a shell that lies still can still be part of a living body; the living snail inside breathes, feeds and grows.

10.2 Essential Conditions for Germination of a Seed

DEFINITION

When a seed turns into a sprout, it has **germinated**.

Activity 10.2: four pots of garden soil with bean seeds, kept 15 days (observe 7–10 days):

Pot	Condition	Result and reason
A	No water, direct sunlight	No germination — **no water**
B	Excess water always above soil, sunlight	No germination — excess water leaves **no air** for the seeds
C	Moderately moist soil, kept in dark	**Germinates** — light is not essential for beans
D	Moderately moist soil, sunlight	**Germinates** — has both water and air

- **Water** softens the seed coat and lets the tiny **embryo** grow into a plant.
- **Air** is used from the spaces between soil particles; these spaces also let roots grow easily.
- **Light/dark:** most seeds do not need light to germinate, but sunlight is needed for the **seedling's further growth**. Coleus and Petunia seeds **need light** (do not cover with soil); Calendula and Zinnia seeds **need darkness** (cover with soil).
- Conclusion: germination of bean seeds requires **the right amount of water and air**.
- Knowledge use: store grains and pulses **dry and away from moisture** so they do not germinate or spoil.

10.3 Growth and Movement in Plants

Activity 10.3: seedlings of bean/gram are fixed on blotting paper on glass plates in beakers A (upright, light from all sides), B (inverted: shoot down, root up) and C (upright, light from one side through a hole in a box).

Set-up	Observation
A: upright, light from all sides	Root grows **downwards**, shoot grows **upwards**
B: inverted	Root **bends and grows downwards**; shoot **bends and grows upwards**
C: light from one direction	Shoot grows **towards the light**; root continues to grow downwards

REMEMBER

Shoots grow upward and towards sunlight; roots grow downward.

▸ Know a scientist — Jagadish Chandra Bose (1858–1937)

Built the **crescograph** to record how plants respond to stimuli such as light, heat, electricity and gravity and to measure their growth rate; showed that plants can **sense and respond to stimuli**.

10.4 Life Cycle of a Plant

Activity 10.4: sow a bean seed and record changes over three months.

Stage I – Seed
Sown in suitable soil.

Stage II – Germination
Seed sprouts; root and shoot appear.

Stage III – Leaves
Young plant grows leaves.

Stage IV – Flowers
Plant matures and flowers appear.

Stage V – Fruit
Flowers turn into fruits (pods) containing seeds.

Death
After fruiting the plant turns yellow, dries and dies; seeds give new plants.

DEFINITION

The whole process from a **seed to a plant and then to the next generation of seeds** is the **life cycle of a plant**.

10.5 Life Cycle of Animals

An animal's life cycle: newborn → growth and development → adult → death; reproduction maintains continuity. Young ones of animals differ in form (e.g. tadpole, larva).

▸ 10.5.1 Mosquito

- Female mosquitoes are blood-sucking insects that **transmit malaria, dengue and chikungunya**. They lay eggs on or near **stagnant water** (desert coolers, planted pots, open containers) so water should not be allowed to stagnate.
- **Four stages:** Egg → Larva → Pupa → Adult. Adult emerges from pupa, rests briefly on water surface and flies; it lives about 10–15 days.
- Larvae and pupae live in water but need air, so they **come to the surface** to breathe.
- **Kerosene oil** sprayed on stagnant water forms a thin layer that blocks air, so larvae and pupae die — breaking the life cycle.
- Activity 10.5: to find which comes first (larva or pupa) separate a few of each, observe daily — larvae turn into pupae, pupae into adults; so larva precedes pupa.
- Body shape differs greatly in each stage.
- **Silk moth** also has egg, larva, pupa and adult stages; larvae spin thread around themselves before becoming pupae, and this fibre is used to make silk. KVIC has set up silk production centres.

▸ 10.5.2 Frog

A frog's eggs are laid in water as a white jelly-like cluster called **spawn**.

Stage	Description	Approx. time
I A – Spawn/eggs	Cluster of eggs in water	Day 1
I B – Embryo	Developing inside the egg	Day 3–4

Stage	Description	Approx. time
II A – Tadpole with tail	Swims with tail, no legs; lives in water	Day 7–10
II B – Tadpole with legs	Hind legs develop but tail present	8–10 weeks
III – Froglet	Looks like a little frog; tail shrinking; spends time on land and water	12 weeks
IV – Adult frog	Tail gone, strong legs for jumping; lives on land and in water	14 weeks

- The tadpole's **tail helps it swim**; it disappears as it grows into a frog.
- Habitat changes: water in the early stages, land and water in the adult.
- Mosquito and frog show **significant changes** in body shape, structure and sometimes habitat during their life cycle.

Feature	Plant life cycle	Animal life cycle
Start	Seed (germination)	Egg or newborn
Growth	Roots, shoots, leaves, flowers, fruits	Larva/young stages to adult
Reproduction	Seeds in fruits give new plants	Eggs/young ones produced
Common	Both pass through birth, growth, reproduction and death; reproduction ensures continuity of life	

10.6 Likely Exam Questions

Likely question	Model answer
List the characteristics of living beings.	Movement, nutrition, growth, respiration, excretion, response to stimuli, reproduction and death.
Do plants move? Give examples.	Yes, without changing place: flowers open, Drosera traps insects, climbers wind around supports, shoots bend towards light.
What is a stimulus? Give an example of a plant response.	Anything that prompts a response; touch makes the touch-me-not fold its leaves.
How do plants respire and excrete?	Through stomata; droplets of excess water and minerals on leaves.
Why is a seed living?	It germinates and shows life processes when conditions are right.
What conditions are needed for seed germination?	Right amount of water and air; light not needed for most seeds.
Why did seeds in excess water not germinate?	Air could not reach them.
What does the bean seedling experiment show about roots and shoots?	Roots grow down and shoots up and towards light, even if the seedling is inverted.
Define life cycle of a plant.	Seed to plant to flowers and fruits with seeds for the next generation.

Likely question	Model answer
Name the stages of a mosquito and a frog's life cycle.	Mosquito: egg, larva, pupa, adult. Frog: egg (spawn), tadpole, froglet, adult.
Why is kerosene sprayed on stagnant water?	It forms a layer blocking air so larvae and pupae die.
What is the advantage of tail in tadpole?	It helps swimming in water.

★ KEY TAKEAWAYS

- Living beings move, need food, grow, respire, excrete, respond to stimuli, reproduce and die.
- Seeds need suitable water and air to germinate; light is needed for later growth.
- Roots grow down, shoots grow up and towards light.
- Plant life cycle: seed → germination → leaves → flowers → fruits/seeds.
- Mosquito: egg-larva-pupa-adult; frog: egg-tadpole-froglet-adult.
- Prevent mosquito breeding by avoiding stagnant water.

CHAPTER 11

Nature's Treasures

*Bhoomi and Surya spend their vacation at Ajji's village in the Western Ghats and learn that fresh air, water, sunlight, forests, soil, rocks, minerals and fuels are **treasures of nature** that make life possible. The chapter describes each resource, how it is used and conserved, and classifies **natural resources** as renewable or non-renewable. Gandhiji's words: Earth provides enough to satisfy every man's need but not every man's greed.*

Big Questions

- Which natural resources do we depend on?
- How can we conserve water and forests?
- What are fossil fuels and why must they be saved?
- What are renewable and non-renewable resources?

11.1 Air

- We need **oxygen** from the air for survival: we can live without food or water for days but not without oxygen for even a few minutes. Most living beings need oxygen.
- Deep breathing (Activity 11.1) brings more fresh air into the lungs; never hold the breath till uncomfortable.

Gas in air	Percentage
Nitrogen	78 %
Oxygen	21 %
Argon, carbon dioxide and other gases	1 %

Percentage means 'parts in 100'. Air is a **mixture of gases**.

- **Moving air is wind** — fast in a storm, gentle as a breeze. Wind rustles leaves, sways clothes, rotates a **firki (paper pinwheel)**.
- A **windmill** works like a firki: wind turns its wings to run flour mills, lift water from wells or **generate electricity**.
- A **windmill farm** has many windmills. Leading Indian wind farms: **Muppandal (Tamil Nadu), Jaisalmer Wind Park (Rajasthan), Brahmanvel (Maharashtra)**.

11.2 Water

- Uses: drinking, cooking, bathing, washing, growing crops, industry.
- Water covers about **two-thirds of Earth's surface**; most is in **oceans and seas and is saline (salty)** — unfit for domestic, agricultural or industrial use.
- **Freshwater** exists in ice sheets and snow, rivers and lakes, and underground. Ice, snow and groundwater are hard to access; only a **very small fraction** in ponds, rivers, lakes and wells is easily accessible.
- Everyone does not have equal access; in some places people walk long distances for drinking water.

▸ Conserving water and preventing pollution

- Switch off taps when not in use, fix leaks, water plants carefully, **recycle water** and **harvest rainwater**.
- **Pollution:** trash, plastic bags and wrappers, and waste from homes and industries dumped into rivers and lakes make water unfit for use.
- **Rainwater harvesting** — collecting and storing rainwater for later use. An age-old Indian practice: **stepwells** — **Bawadi in Rajasthan, Vav in Gujarat** (e.g. Toorji ka Jhalra, Jodhpur) store rainwater and water seeping from lakes, ponds and rivers through stone-lined trenches.
- **World Water Day** is observed on **22 March**.

11.3 Energy from the Sun

> **IMPORTANT**
> The **Sun is the main source of energy on Earth** — it provides heat and light. Plants use sunlight to make food; animals eat plants (and other animals), so all living beings depend on the Sun. The cow grazing grass gets energy from the Sun *through the grass*, not from standing in sunshine.

- Heat uses: drying clothes, chillies, raw mangoes; light for photosynthesis and vision.
- **Solar panels** capture the Sun's energy and produce electricity (rooftops, street lights, traffic signals). **Solar cooker** cooks food; **solar water heater** heats water.
- Without sunlight for days we would need artificial lighting and heating; solar panels would not work.

11.4 Forests

- Forests are large areas with **dense growth of herbs, shrubs and trees**; a natural home of wild animals, birds and insects, giving food and shelter.
- Forest products: wood, fruits, honey, medicinal herbs, gum, bamboo, lac.
- Forest cover is shrinking mainly because of **large-scale tree cutting**; regrowing a forest takes many years, so use forests responsibly.
- Village tradition: do not pluck fruits from trees; collect only fallen ones (nellikai) so animals and birds can eat.
- Roots **hold the soil** and prevent it from being washed away; fallen leaves decay and **enrich the soil** — an example of **recycling in nature**.
- **Van Mahotsav**: week-long forest festival in July when new trees are planted to increase green cover.
- **Chipko movement** (early 1970s, Uttarakhand): local women hugged trees to prevent them from being felled.
- Sacred groves are another Indian tradition of protecting forests.

11.5 Soil, Rocks and Minerals

▸ Soil

- Soil contains sand, pebbles, roots, insects, worms, tiny organisms and decayed plants and animals; colour and texture differ from place to place (Activity 11.4 — wash hands after handling soil).
- **Earthworms** turn and loosen the soil; spaces between particles give air and room for roots.
- Soil is formed by **disintegration of rocks** by the Sun, water and living organisms over **thousands of years**.
- Different soils suit growing plants or making bricks. Soil supports **biodiversity**.

▸ Rocks and minerals

- Rocks are used to build houses, temples, roads, dams, table tops; slate for roofing, laterite as bricks; granite, sandstone and marble are important rocks. Early humans made **hand axes and arrowheads** from stone.
- Rocks are made of **minerals**; metals like aluminium, gold, copper and iron are extracted from minerals. Minerals are used in aircraft, cars, jewellery, cosmetics, electronics. A basic mobile phone contains about a dozen minerals (gold, silver, copper, cobalt etc.).
- Rocks take thousands to millions of years to form — conserve them.

11.6 Fossil Fuels

DEFINITION

Fossil fuels — **petroleum, natural gas and coal** — formed from the remains of microorganisms and plants buried deep inside the earth, over **millions of years**.

Fuel	Obtained / used for
Petrol, diesel, kerosene	Obtained from petroleum; fuel for vehicles (petrol for scooters/bikes)
Natural gas	Cooking, electricity generation; as **CNG** (Compressed Natural Gas) for vehicles — cleaner than petrol/diesel
Coal	Mainly for producing electricity; found in several parts of India
LPG	Domestic cooking fuel replacing coal, wood and dung cakes

- Fossil fuels are limited and will run out if used at present rates.
- Burning gives **smoke and carbon dioxide**, causing air pollution.
- Conserve by **walking or cycling short distances**, using **public transport**, sharing vehicles; use electric vehicles, which emit no smoke, and other alternate energy sources.

11.7 Natural Resources: Renewable and Non-renewable

Type	Meaning	Examples
Natural resources	Resources obtained from nature	Air, water, sunlight, forests, soil, rocks, minerals, fossil fuels, plants, animals
Human-made resources	Created by humans using natural resources	Electric bulb, furniture, solar panels, bicycles
Renewable	Renewed/replenished by nature within a reasonable time	Air, water, forests (also sunlight, wind)
Non-renewable	Limited; not replenished within a reasonable time	Coal, petroleum, natural gas, minerals, rocks, soil

REMEMBER

'Renewable' does not mean unlimited — use water, forests and air judiciously. Soil is listed as non-renewable because it forms over thousands of years.

11.8 Resources We Use

Daily activities use resources: washing clothes (water), making clay toys (soil), collecting firewood (forest), making kites (paper from trees), breakfast (plants/animals, water, fuel). Conserve and use resources responsibly to meet present needs without harming the future or the environment.

11.9 Likely Exam Questions

Likely question	Model answer
Name the gases in air with their percentages.	Nitrogen 78%, oxygen 21%, argon, carbon dioxide and others 1%.
What is wind? What is a windmill farm?	Moving air; a place with many windmills generating electricity, e.g. Muppandal.
Why is most water of Earth not usable?	It is saline ocean water; usable freshwater is a very small fraction.
List ways of saving water.	Close taps, fix leaks, recycle water, harvest rainwater.
What is rainwater harvesting? Give a traditional example.	Collecting rainwater for use; stepwells (Bawadi/Vav).
Why is the Sun the main source of energy?	Plants use sunlight to make food and animals depend on plants; it gives heat and light.
State uses of solar energy.	Solar panels for electricity, solar cooker, solar water heater, drying.
Why should forests be conserved?	They shelter wildlife, hold soil, enrich it and give products; regrowing takes long.
What was the Chipko movement?	Women in Uttarakhand hugged trees to stop their felling.
How is soil formed?	Rocks break down by Sun, water and organisms over thousands of years.
What are fossil fuels? Why non-renewable?	Petroleum, coal, natural gas formed over millions of years; limited and not quickly replaced.
Differentiate renewable and non-renewable resources.	Renewable are replenished in reasonable time (air, water); non-renewable are not (coal, minerals).

★ KEY TAKEAWAYS

- Air (78% nitrogen, 21% oxygen), water, Sun, forests, soil, rocks, minerals and fossil fuels are natural resources.
- Freshwater is limited — save and harvest it; keep it pollution-free.
- The Sun is the main source of energy.
- Forests protect soil and biodiversity; Chipko and Van Mahotsav promote protection.
- Fossil fuels (coal, petroleum, natural gas) are non-renewable and pollute the air when burnt.
- Renewable resources can be replenished; use all resources judiciously.

CHAPTER 12

Beyond Earth

*Yangdol and Dorjay in Nubra, Ladakh, watch a star-filled night sky free of air and light pollution. The chapter explains **stars and constellations**, how to watch the night sky and find the **Pole Star**, the members of the **Solar System** (Sun, planets, moons, asteroids, comets), the **Milky Way** galaxy and the Universe.*

Big Questions

- What are constellations and how are they useful?
- How can we find the Pole Star and Orion?
- What does the Solar System contain?
- What lies beyond the Solar System?

12.1 Stars and Constellations

- Stars **shine with their own light**; some look bright, some dim. Ancient people imagined patterns joining stars into animals, objects or story characters; patterns differ in different cultures.
- These patterns helped to **recognise stars and find directions** — at sea or land, even before the magnetic compass; still a backup in emergencies.
- A **constellation** was originally a group of stars forming a pattern; now it means a **defined region of the sky**. The **International Astronomical Union (IAU)** fixed boundaries in the early 20th century — **88 constellations** divide the entire sky.

Constellation / star	Details
Orion (hunter)	Three stars in a line in the middle form the hunter's belt; seen with dog **Canis Major** and bull **Taurus**
Sirius	In Canis Major; the **brightest star** in the night sky
Betelgeuse (Ārdrā), Aldebaran (Rohini), Pleiades (Krittika)	Indian *nakshatras*: Ardra in Orion; Krittika (Pleiades) and Rohini (Aldebaran) in Taurus
Big Dipper	Part of **Ursa Major**; called **Saptarishi** in India
Little Dipper	Part of **Ursa Minor**; contains the **Pole Star (Polaris, Dhruva tara)**
Pole Star	Appears nearly **stationary in the north**; helps find north in the Northern Hemisphere; not very bright; **not visible from the Southern Hemisphere**

Stories: tribes of Central India see four Big Dipper stars as a 'grandmother's cot' and three as thieves; Konkan fishermen see a boat.

12.2 Night Sky Watching

- **Light pollution** — excessive artificial light at night — along with smoke and dust, hides stars in cities; more stars are visible in villages. Tall buildings and trees also block the view. Dark-sky reserves and parks protect dark skies for research.
- Not all constellations are visible from every place or on all nights.

- Tools: sky-mapping apps (**Sky Map, Stellarium**), printed charts, a compass, notebook; telescopes and binoculars show dim objects brighter.

Choose place
A dark, open area away from lights, tall buildings and trees, with adult supervision (never go alone).

Choose time
Use an app to select date/time; pick a night without Moon or clouds, especially to spot the faint Pole Star.

Carry
Sky map or printouts, magnetic compass, notebook.

Adjust eyes
Wait about half an hour for your eyes to adapt to the darkness.

- **Finding the Pole Star:** locate the Big Dipper (look north in summer around 9 p.m.). Imagine a line through the two stars at the end of its cup towards the north; at about **five times** the distance between those two stars lies the Pole Star.
- **Finding Orion:** best seen from India in **December to April** after sunset; find the three bright stars in a short row (belt); extending that line towards the **east** leads to **Sirius**.

12.3 Our Solar System

▸ The Sun

- A **star**, the nearest to us, an extremely hot ball of gases; produces heat and light; main source of energy for Earth — plants, food, oxygen, climate, seasons, weather, winds and the water cycle depend on it. In India the Sun is worshipped as **Surya**.
- About **100 times** the Earth's diameter; seems small because of distance (~**150 million km**). One **astronomical unit (au)** is about the Sun–Earth distance.
- Other stars are far away, so look like points of light; many are bigger than the Sun. Stars are invisible by day because of the Sun's brightness.
- Nearest star after the Sun: **Proxima Centauri**, about **269,000 au** away.
- The Sun is the **largest and heaviest** object in the Solar System and gives almost all its energy; other objects shine by **reflected sunlight**.

▸ Planets

DEFINITION

A **planet** is a large, nearly spherical object that **revolves around the Sun**. **Revolution** is the movement of an object around the Sun (or another body). Earth takes **about one year** to revolve and rotates on its axis in about **24 hours** (one day).

Order from Sun	Planet (Indian name)	Notes
1	Mercury (Budha)	Inner rocky planet
2	Venus (Shukra)	Morning/Evening Star (not a star); brightest after Sun and Moon; hotter than Mercury because its atmosphere traps heat
3	Earth (Prithvi)	Blue Planet — water covers most of the surface
4	Mars (Mangala)	Red Planet — reddish soil; two moons

Order from Sun	Planet (Indian name)	Notes
5	Jupiter (Brihaspati/Guru)	Giant, gas and ice; rings
6	Saturn (Shani)	Giant; prominent rings
7	Uranus	Giant, gas and ice
8	Neptune	Giant, gas and ice

- Memory aid: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune.
- **Inner four** planets are smaller with solid rocky surfaces; **outer four** are much larger, mostly gas and ice, with **ring systems** of dust and rock.
- Planets are colder generally the farther they are from the Sun; atmosphere can change this.
- **Pluto** lies beyond Neptune, smaller than Earth's Moon; in **2006 IAU** redefined 'planet' and Pluto became a **dwarf planet**.
- Mercury, Venus, Mars, Jupiter and Saturn are visible to the naked eye; **stars twinkle**, planets do not twinkle much. Venus can be seen in the east before sunrise or the west after sunset.

▸ Natural satellites and the Moon

- Objects that move around planets are **satellites**; **moons** are natural satellites. Earth has 1 moon, Mars 2; Jupiter, Saturn, Uranus and Neptune have many.
- The **Moon**: revolves around Earth in about **27 days**; about one-quarter of Earth's diameter; about **3,84,000 km** away; has hardly any atmosphere, water or life; **craters** (bowl-shaped) formed by impacts of asteroids/rocks stay for a very long time.
- **Chandrayaan-1** (2008), **Chandrayaan-2** (2019), **Chandrayaan-3** (July 2023) — Vikram lander with Pragyan rover soft-landed near the Moon's **south pole on 23 August 2023**, a first for any country; 23 August is **National Space Day**. **Chandrayaan-4** is planned to bring back samples.

▸ Asteroids and comets

Object	Description
Asteroids	Small, rocky, irregular; most revolve in the **asteroid belt between Mars and Jupiter**; sizes from ~10 m to ~500 km
Comets	Icy-rocky bodies of dust, gas, rock and ice with long **tails**; near the Sun the ice evaporates forming the tail; **Halley's Comet** appears every **76 years** (last in 1986); Sanskrit name *Dhumaketu*

> **IMPORTANT**
> The **Solar System** = Sun + eight planets + their moons + asteroids, comets and other small objects.

12.4 The Milky Way Galaxy

A faint band of light across the sky on moonless nights from dark places is our home galaxy, the **Milky Way (Akasha Ganga)**. A **galaxy** has millions to billions of stars; the Solar System is part of it.

12.5 The Universe

There are many galaxies beyond the Milky Way, and the Universe contains them all. Whether life exists elsewhere is unknown; the search focuses on **exoplanets** — planets around other stars — and no evidence has been found yet.

12.6 Likely Exam Questions

Likely question	Model answer
What is a constellation? How many are there?	A defined region of the sky with a star pattern; 88 officially.
How were star patterns useful to ancient people?	To identify stars and find direction at sea and on land.
How do you locate the Pole Star?	Extend a line through the two end stars of the Big Dipper's cup about five times their separation.
Why is the Pole Star important?	It appears nearly fixed in the north, showing north direction.
What is light pollution?	Excess artificial light at night that hides stars.
Why can't we see stars in the daytime?	The Sun's brightness outshines them.
Why does the Sun look bigger than other stars?	It is much closer to us.
Define planet and revolution.	Large nearly spherical body revolving around the Sun; movement around the Sun.
Name the planets in order from the Sun.	Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune.
Why is Mars called the Red Planet and Earth the Blue Planet?	Mars has reddish soil; Earth's surface is mostly water.
Why is Pluto not a planet?	IAU redefined planet in 2006; Pluto is a dwarf planet.
What are asteroids and comets?	Rocky irregular bodies in the asteroid belt; icy bodies with tails near the Sun.
Write about Chandrayaan-3.	Launched July 2023; Vikram lander and Pragyan rover landed near Moon's south pole on 23 August 2023, celebrated as National Space Day.

★ KEY TAKEAWAYS

- Constellations are star patterns/regions (88 in total); Orion, Big Dipper, Little Dipper; the Pole Star shows north.
- Light pollution reduces visibility of stars; watch the sky from dark, open places.
- The Sun is a star and the main energy source; the Solar System has eight planets, moons, asteroids and comets.
- Planets in order: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune.
- The Moon is Earth's natural satellite and revolves in about 27 days; India's Chandrayaan-3 landed near the south pole.
- The Solar System belongs to the Milky Way galaxy within the Universe.